

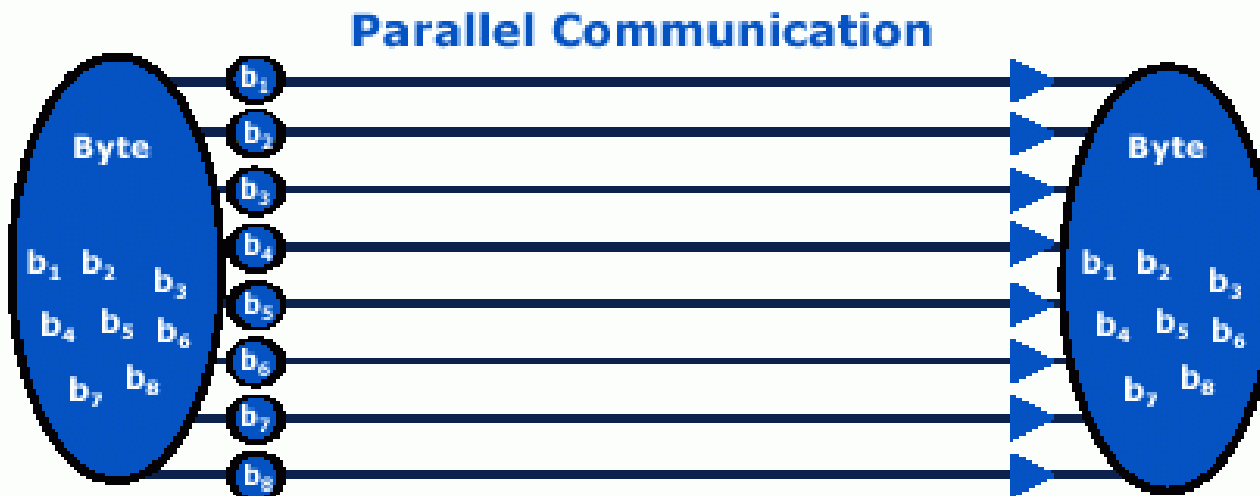
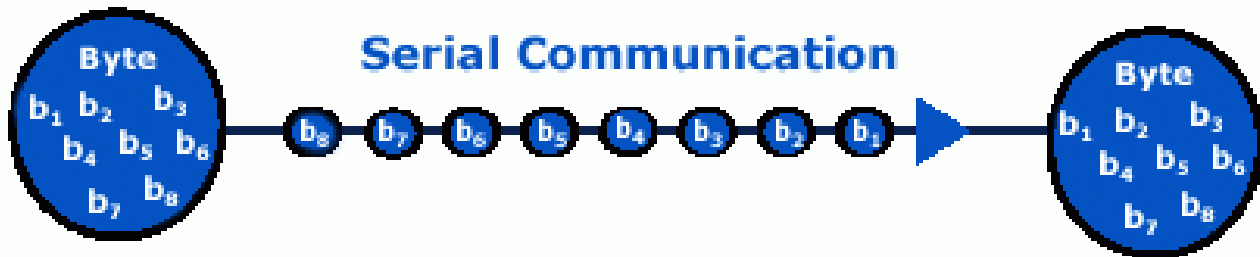
Physical Layer

- Transmission Modes
- **Modulation**
- Transmission Impairment
- Multiplexing
- Switching modes
- Transmission medium

Transmission Modes

Serial vs. parallel communications

- **In serial communications**
 - A single bit will be transferred at a time using the communication channel
 - Bits will be reassembled at the destination
 - Mostly used by computer peripherals like printers
- **In parallel communications**
 - Multiple bits (eg. Eight bits) will be transferred at a time
 - Needs multiple (parallel) communication channels



Serial transmission can be either asynchronous or synchronous.

■ **Asynchronous Serial Transmission**

- It is named so because there is no importance of timing.
- Data-bits have specific pattern and they help receiver recognize the start and end data bits.
- For example, a 0 is prefixed on every data byte and one or more 1s are added at the end.
- Two continuous data-frames (bytes) may have a gap between them.

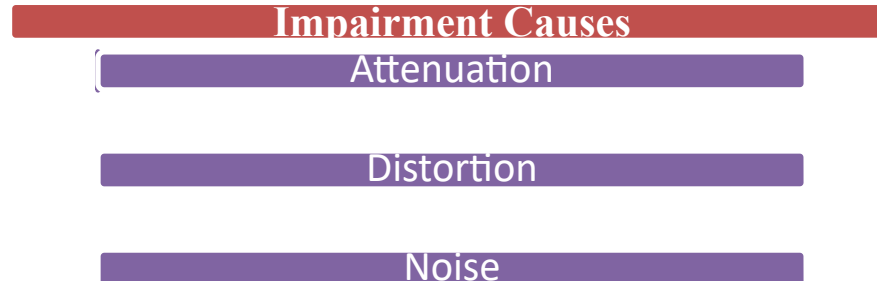
Serial transmission can be either asynchronous or synchronous.

■ Synchronous Serial Transmission

- Timing in synchronous transmission has importance as there is no mechanism followed to recognize start and end data bits.
- There is no pattern or prefix/suffix method. Data bits are sent in burst mode without maintaining gap between bytes (8- bits). Single burst of data bits may contain a number of bytes. Therefore, timing becomes very important.
- It is up to the receiver to recognize and separate bits into bytes. *The advantage of synchronous transmission is high speed, and it has no overhead of extra header and footer bits as in asynchronous transmission.*

Transmission Impairment

- Signals travel through transmission media, which are **not perfect**. The imperfection **causes signal impairment**.
- This means that the signal at the **beginning of the medium** is not the **same as the signal at the end of the medium**. What is **sent** is not what is **received**.
- Three causes of impairment are **attenuation, distortion, and noise**.

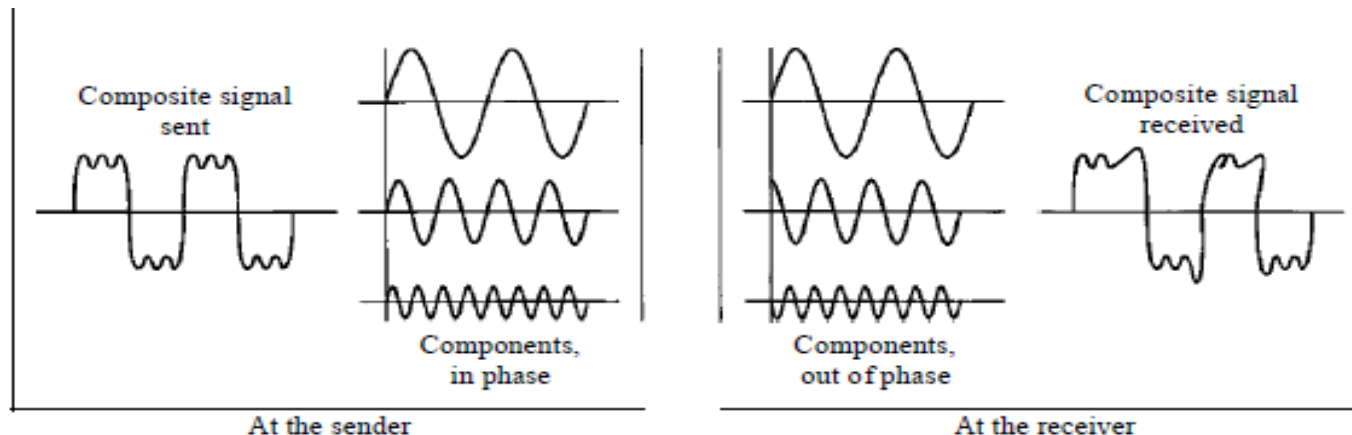


Attenuation

- Attenuation means a **loss of energy**.
- When a signal, travels through a medium, it loses some of its energy in overcoming the resistance of the medium.
- That is why a wire carrying electric signals gets warm, if not hot, after a while. **Some of the electrical energy in the signal is converted to heat.**
 - To compensate for this loss, amplifiers are used to amplify the signal.
 - Attenuation is measured in terms of Decibels.
 - The decibel (dB) measures the relative strengths of two signals or one signal at two different points.
 - Note that the decibel is negative if a signal is attenuated and positive if a signal is amplified.

Distortion:

- Distortion means that the signal **changes its form or shape**.
- Distortion can occur in a signal made of **different frequencies**.
- Each signal component has its own **propagation speed** through a medium and, therefore, **its own delay** in arriving at the final destination.
- **Differences in delay may create a difference in phase if the delay is not exactly the same as the period duration.**



Noise

- Noise is another cause of impairment. Several types of noise, such as **Thermal noise, Induced noise, Crosstalk, and Impulse noise**, may corrupt the signal.
- **Thermal noise** is the random motion of electrons in a wire which creates an extra signal not originally sent by the transmitter.
- **Induced noise** comes from sources such as **motors and electrical device**. These devices act as a sending antenna, and the transmission medium acts as the receiving antenna.
- **Crosstalk** is called electromagnetic interference (EMI). It **is the effect of one wire on the other**. One wire acts as a sending antenna and the other as the receiving antenna.
- **Impulse** noise is **a spike** (a signal with high energy in a very short time) that comes from power lines, lightning, and so on.

Reading Assignment

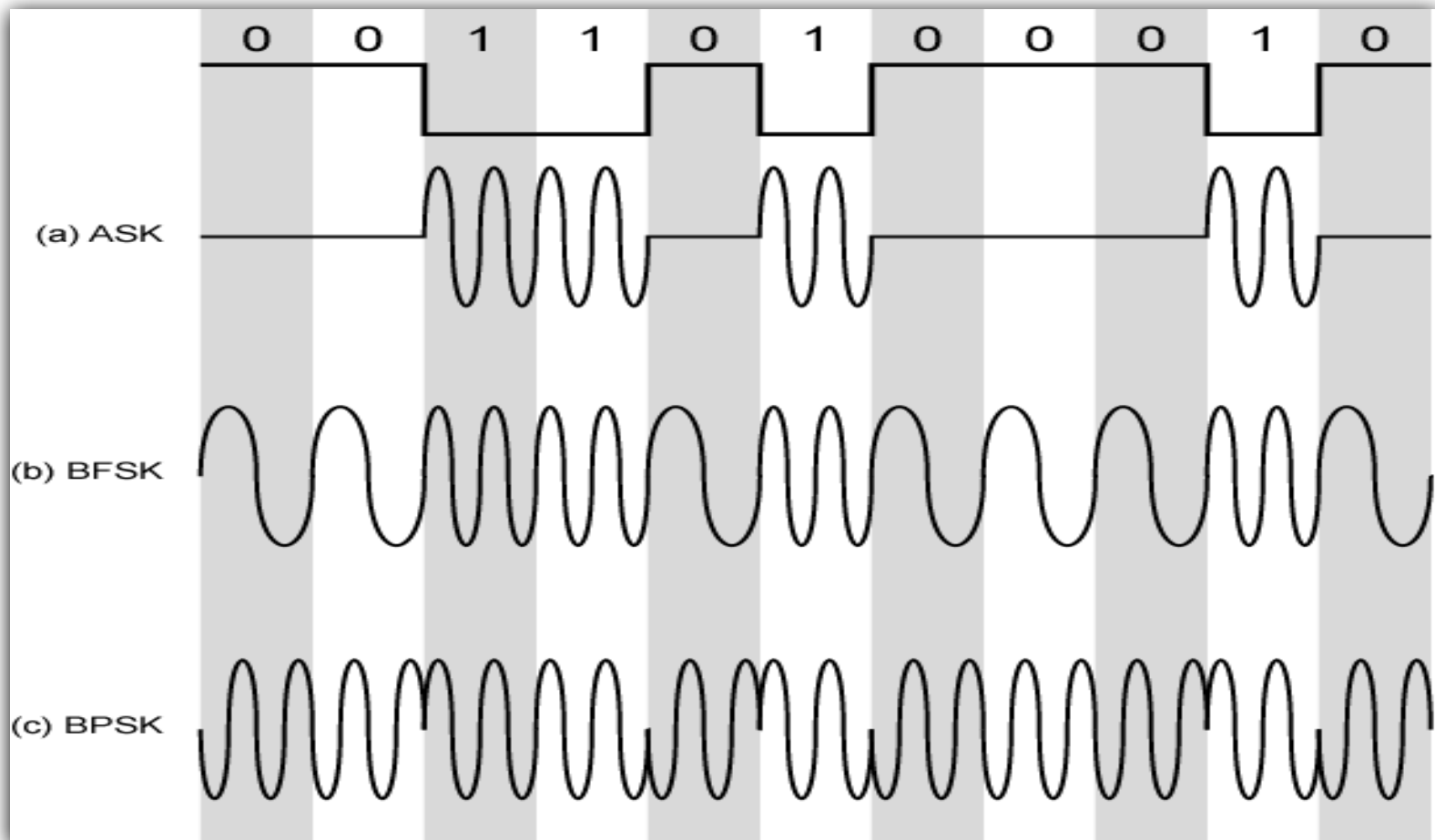
MODULATION /DATA ENCODING

Modulation

- **Modulation** is the process of transforming a carrier signal so that it can carry the information of a message signal.
- It superimposes the contents of the message signal over a high-frequency carrier signal, which is then transmitted over communication channels.
- Modulation can be of two types –
 - Analog Modulation
 - Digital Modulation

Digital Data to Analog Signal Encoding Techniques

- Modulation involves operation on one or more of the three characteristics of a carrier signal: **amplitude, frequency, and phase.**
- Accordingly, there are three basic encoding or modulation techniques for transforming digital data into analog signals
 1. Amplitude-shift keying (ASK)
 2. Frequency-shift keying (FSK)
 3. Phase-shift keying (PSK)



Analog Modulation

- Here, the analog information signal is transformed to the analog carrier signal so that it can travel large distances without substantial loss.
- Analog modulation can be of three types –
 - Amplitude Modulation
 - Frequency Modulation
 - Phase Modulation

Digital Modulation

- Digital modulation is the process of converting a digital bit stream into an analog carrier wave for transmission via a communication channel.
- Digital modulation is broadly divided into two categories –
 - Bandpass Modulation as in baseband transmission:**
 - Here, the bits are converted directly into signals.
 - Passband Modulation as in passband transmission:**
 - Here, the amplitude, phase or frequency of the carrier signal is regulated to transmit the bits.

Digital Modulation

Baseband Modulation

NRZ(Non
return to
zero)

NRZI(NRZ
invert)

Mancheste
r Encoding

Passband modulation

ASK(A
mplitu
de Shift
Keying)

FSK(Fre
quency
Shift
Keying)

PSK(Ph
ase
Shift
Keying)

BPSK(Bi
nary
PSK)

DIGITAL TRANSMISSION

- how to convert digital data into digital signals ?
- It can be done in two ways:
 - line coding
 - block coding.
- For all communications, line coding is necessary whereas block coding is optional.

Digital-to-Digital Conversion

line coding

- Uni-polar
- Bipolar
- Polar :
 - 4 types
 - » Polar Non Return to Zero (Polar NRZ)
 - » Return to Zero (RZ)
 - » Manchester
 - » Differential Manchester

block coding

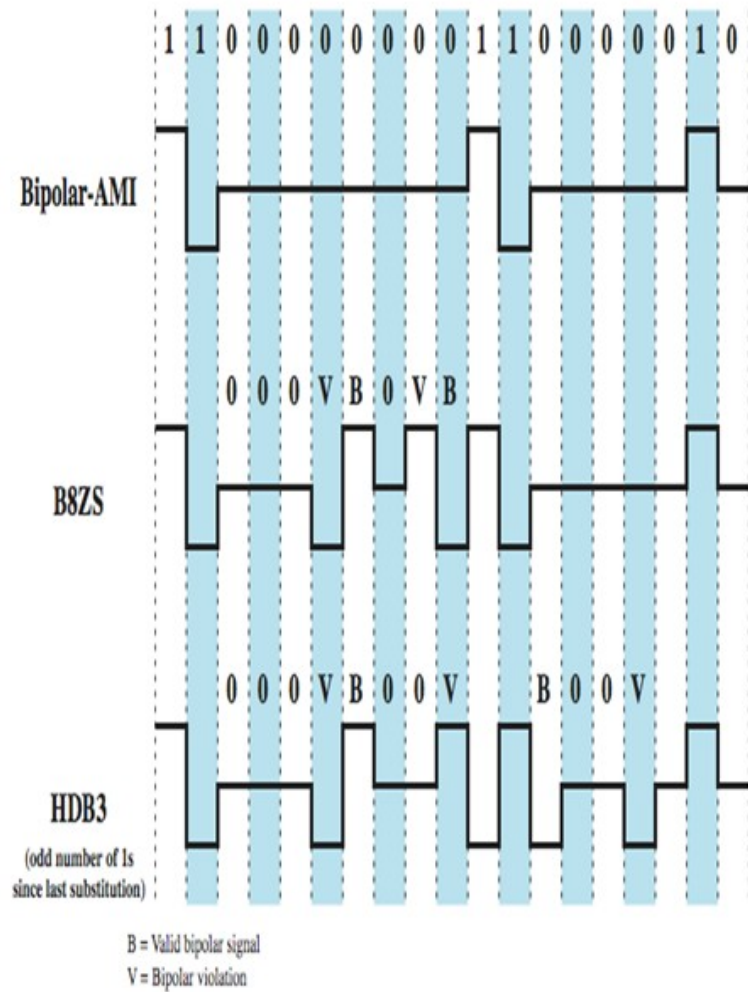
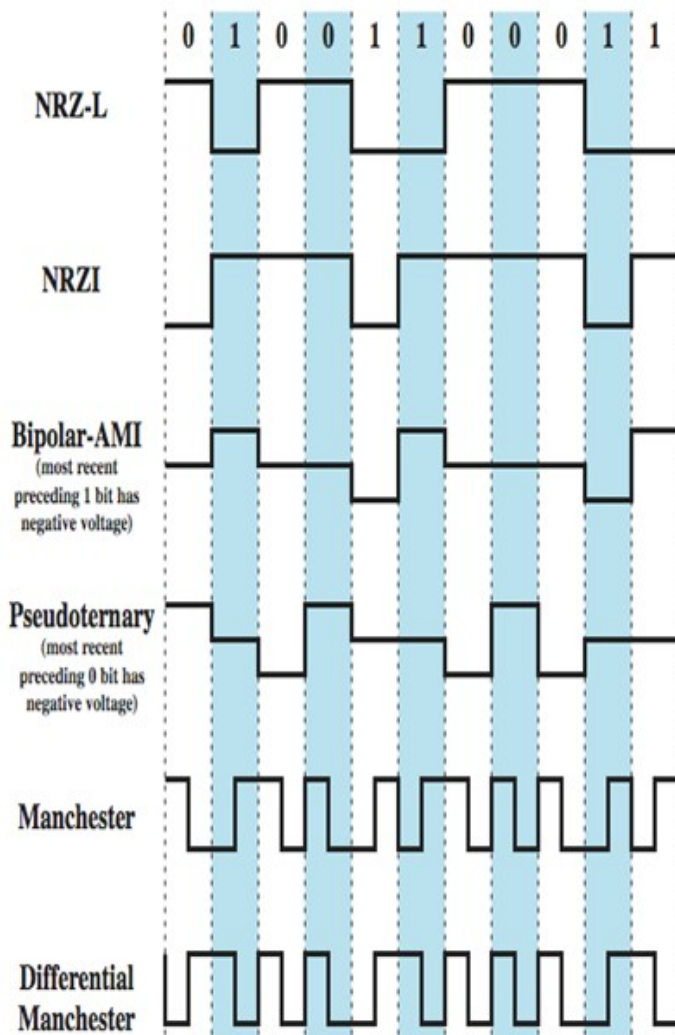
- involves three steps:
 - Division
 - Substitution
 - Combination.

Analog-to-Digital Conversion

- Analog data is a continuous stream of data in the wave form whereas digital data is discrete. To convert analog wave into digital data, we use Pulse Code Modulation (PCM).
- PCM is one of the most commonly used method to convert analog data into digital form. It involves three steps:
 - Sampling
 - Quantization
 - Encoding.

Assignment #1

- Describe the following digital signal encoding formats (which format is used by Ethernet?)
 1. **Non return-to-Zero-Level (NRZ-L)**
 2. **Non return to Zero Inverted (NRZI)**
 3. **Bipolar-AMI**
 4. **Pseudoternary**
 5. **Manchester**
 6. **Differential Manchester**



ANALOG TRANSMISSION

- To send the digital data over an analog media, it needs to be converted into analog signal. There can be two cases according to data formatting.
- **Bandpass:** The filters are used to filter and pass frequencies of interest. A bandpass is a band of frequencies which can pass the filter.
- **Low-pass:** Low-pass is a filter that passes low frequencies signals.
- When digital data is converted into a bandpass analog signal, it is called digital-to- analog conversion. When low-pass analog signal is converted into bandpass analog signal, it is called analog-to-analog conversion.

Digital-to-Analog Conversion

- When data from one computer is sent to another via some analog carrier, it is first converted into analog signals. Analog signals are modified to reflect digital data.
- An analog signal is characterized by its amplitude, frequency, and phase.
- There are three kinds of digital-to-analog conversions:
 - Amplitude Shift Keying
 - Frequency Shift Keying
 - Phase Shift Keying

Analog-to-Analog Conversion

- Analog signals are modified to represent analog data. This conversion is also known as **Analog Modulation**.
- Analog modulation is required when bandpass is used.
- Analog to analog conversion can be done in three ways:
 - Amplitude modulation
 - Frequency modulation
 - Phase modulation

MULTIPLEXING

MULTIPLEXING

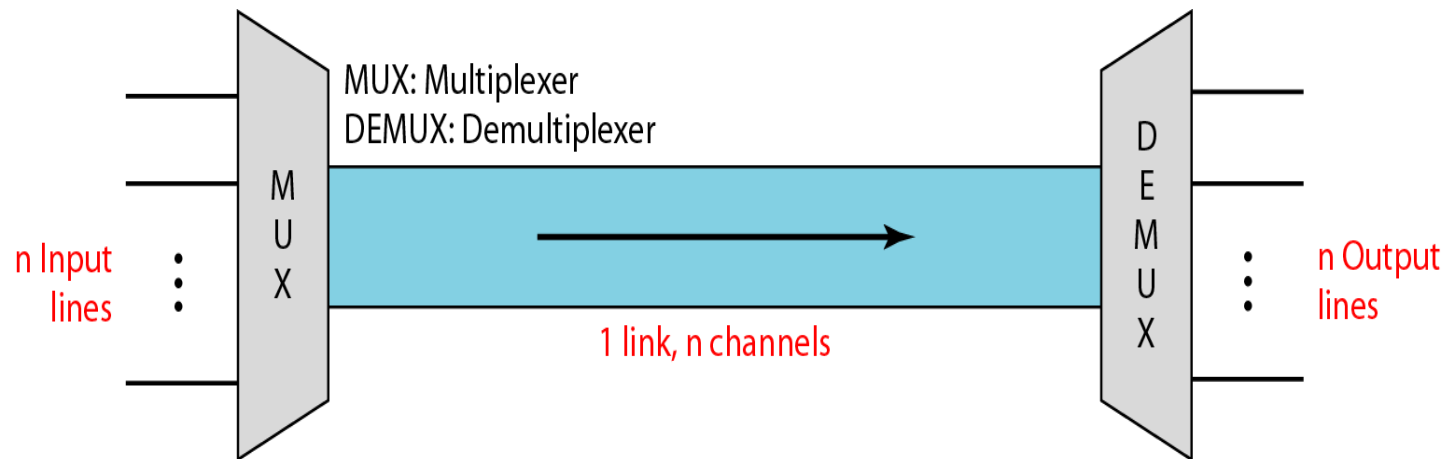
Multiplexing

- In data communication, there might be a need to **share a single media for multiple communication** (media/bandwidth sharing) in order to utilize the available bandwidth wisely to achieve specific goals. Sharing of a single media (fiber, coaxial, microwave,..) is known as **multiplexing**.
- Whenever the bandwidth of a medium linking two devices is greater than the bandwidth needs of the devices, the link can be shared.

Contd.

- This is a method of combining more than one signal over a shared medium.
- **Multiplexing** divides the capacity of a communication channel into several logical channels, each for a data stream.
- The method of extracting the original data streams from the multiplexed signal is called **Demultiplexing**.

Contd.



Types of Multiplexing

Multiplexing

Frequency Division Multiplexing

Time Division Multiplexing

Code Division Multiplexing

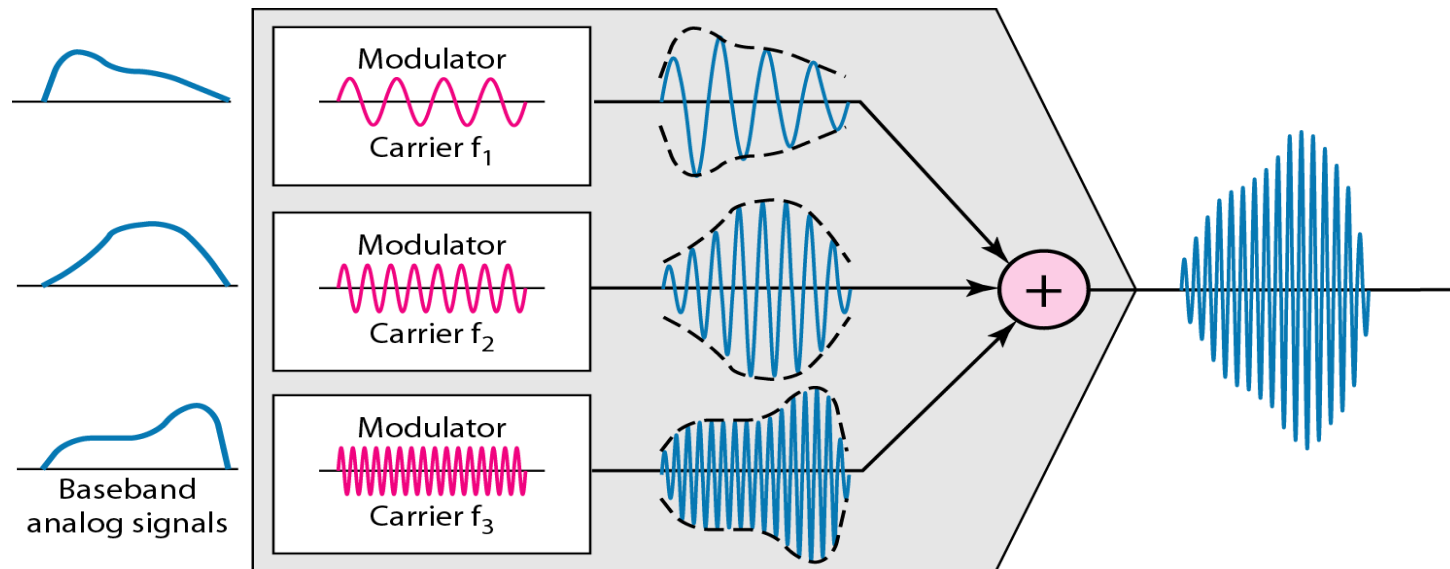
Frequency-division Multiplexing (FDM)

- It is an analog technology.
- When the carrier is frequency ,FDM is used.
- FDM can be used in both **wired and wireless networking** for transmitting **large amounts of data at high speeds**.
- FDM is the **simplest** and **oldest** form of multiplexing in wireless networking technology.
- FDM involves simultaneously **transmitting multiple signals on different frequencies over the same line or channel**.

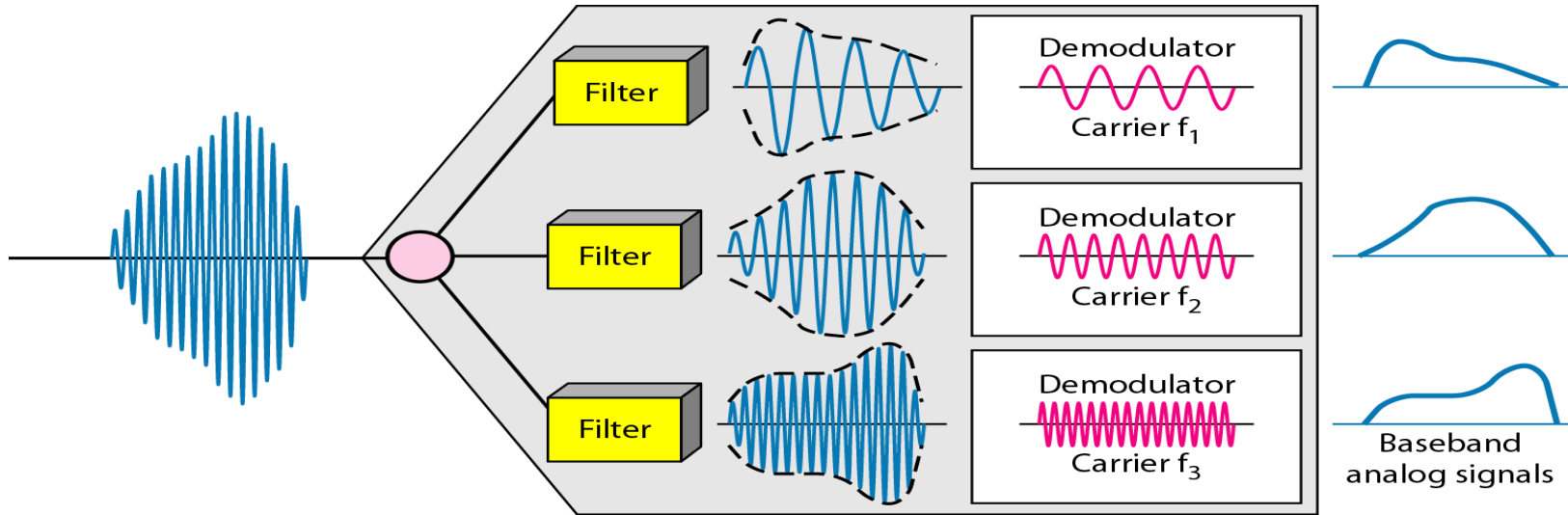
Contd.

- Channels can be separated by strips of unused bandwidth guard bands –to prevent signals from overlapping.
- Signals from different data sources ($f_1, f_2, f_3 \dots$) are fed into a multiplexer that modulates each signal and transmits them at **different frequencies**.
- These signals are then transmitted over the wire or through wireless communication and are separated at the destination into individual data signals using a **demultiplexer**.

FD Multiplexing



FD DEmultiplexing



Application of FDM

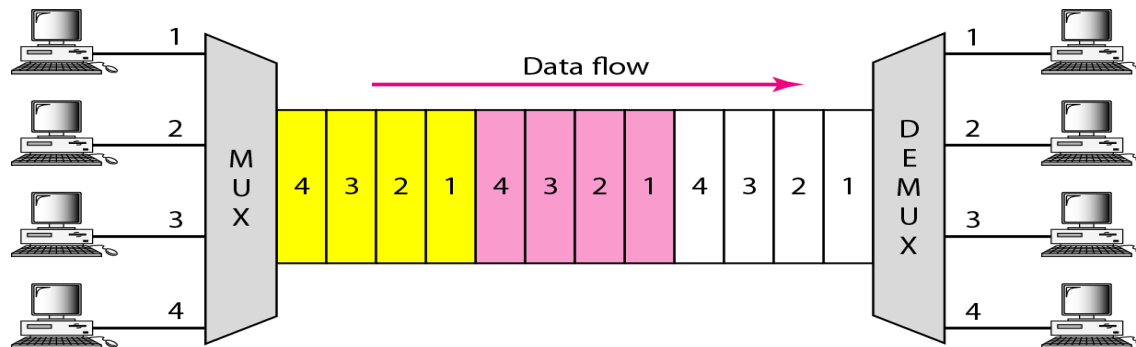
- AM and FM radio broadcasting
- Television broadcasting
- The first generation of cellular telephones

Time-division multiplexing (TDM)

- A digital process for transmitting **multiple data streams** in **a single communication path**.
- **Instead of sharing a portion of the bandwidth as in FDM, time is shared.**
- In TDM, the data from different input channels is divided into **fixed-length segments** and then combined in **round-robin fashion** into a single output data stream, which can then be transmitted over a single channel transmission system and DEmultiplexed at the destination location.

Contd.

- TDM is a digital multiplexing technique for combining **several low-rate digital channels into one high-rate one.**



Contd.

- **One weakness** in TDM is that if an input channel does not have anything important to carry for a time, **empty segments** are inserted into the output stream anyway.
- For example, if channel A is not transmitting data, one-third of the output channel is not being used.
- You can overcome this weakness by using a more sophisticated multiplexing technique called **statistical multiplexing**.

Contd.

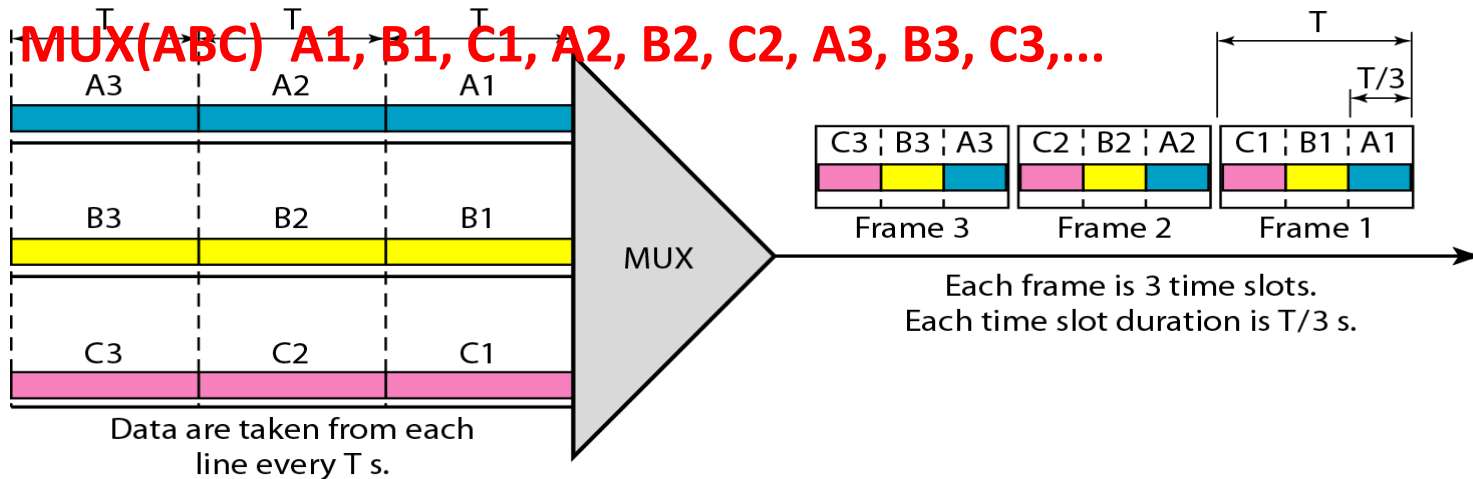
- For example, if input streams A, B, and C are divided into segments as shown here (if we have n input lines then there are n slots in one frame)

A: A1, A2, A3,...

B: B1, B2, B3,...

C: C1, C2, C3,...

the output stream will look like this:



Statistical Multiplexing

- Asynchronous TDM (time slots are not fixed) , for n input lines m slots where $m < n$
- allows information from a number of channels to be combined for transmission over a single channel.
- It dynamically allocates bandwidth to each channel on an as-needed basis. This is in contrast to TDM, in which quiet devices use up a portion of the multiplexed data stream, filling it with empty packets.
- Statistical multiplexing allocates bandwidth only to channels that are currently transmitting.

Statistical Multiplexing

- It packages the data from the active channels into packets and dynamically feeds them into the output channel, usually on a **FIFO (first in, first out) basis**, but it's also able to allocate extra bandwidth to specific input channels.
- Statistical multiplexing is sometimes referred to as **statistical time-division multiplexing (STDM)**

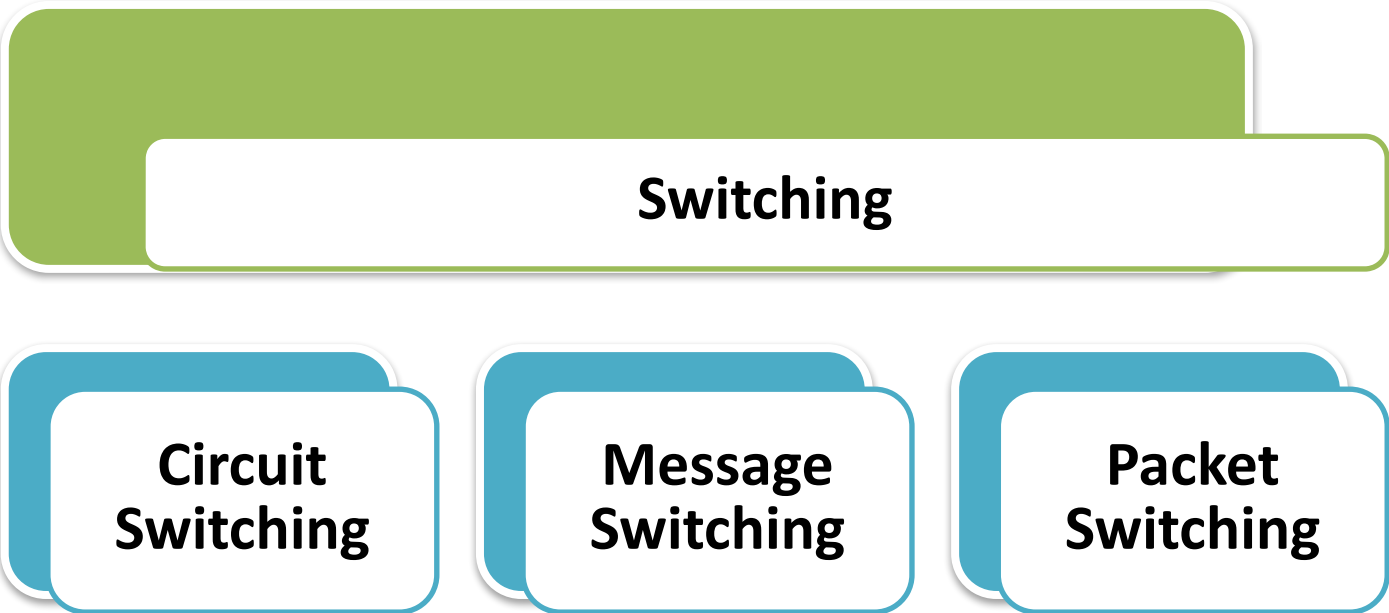
SWITCHING

SWITCHING

Switching

- Switching is a mechanism by which data/information sent from source towards destination which are not directly connected.
- Networks have interconnecting devices, which receives data from directly connected sources, stores data, analyze it and then forwards to the next interconnecting device closest to the destination.

Switching



Switching

**Circuit
Switching**

**Message
Switching**

**Packet
Switching**

Circuit Switching

- When two nodes communicate with each other over a dedicated communication path, it is called circuit switching.
- To transfer data circuit(permanent or temporary) must be established.
- Establish a circuit ,transfer data ,disconnect the circuit
- Designed for voice applications
- Example : **Telephone**

Message Switching

- In the middle of circuit switching and packet switching
- The whole message is treated as a data unit and is switching /transferred in its entirety.
- Store ,buffer ,forward
- Disadvantage :
 - Every switch in transit path needs enough storage to accommodate entire message
 - Very slow
 - Not a solution for streaming media and real-time applications

Packet Switching

- shortcomings of message gave birth to an idea of packet switching.
- Entire message is broken down into smaller chunks called packets.
- The switching information is added in the header of each packet and transmitted independently.
- The internet uses packet switching.

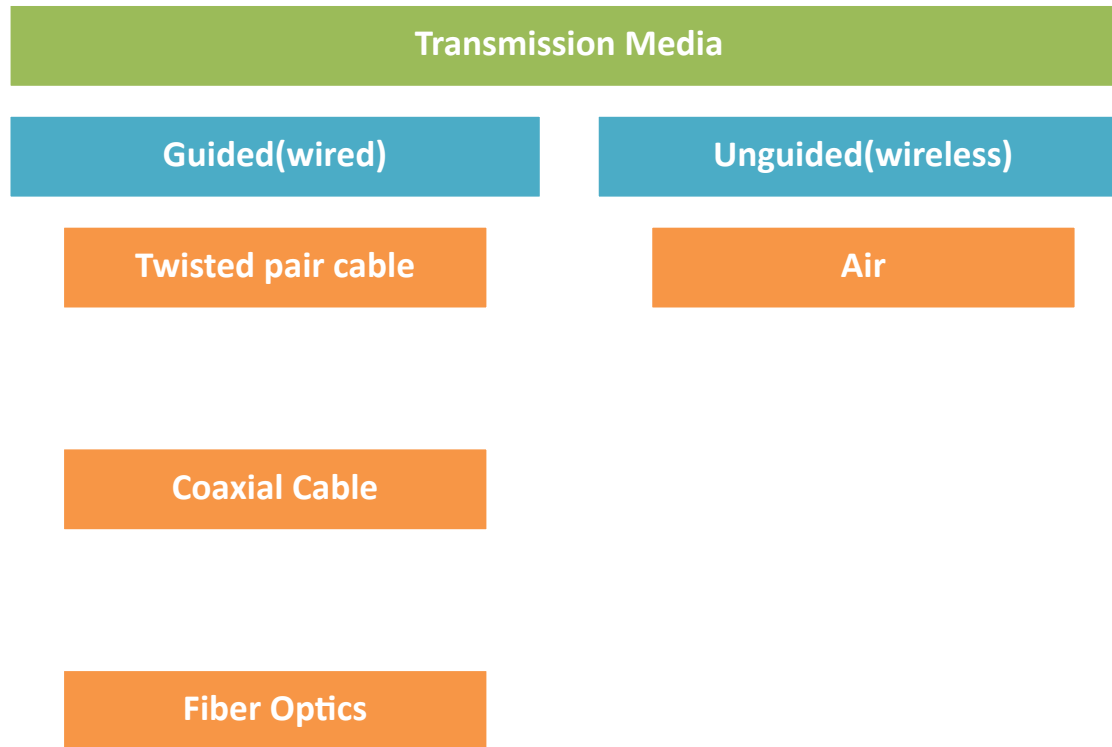
Transmission medium

TRANSMISSION MEDIUM

Transmission medium

- ✧ is the physical path between transmitter and receiver in a data transmission system.
- ✧ Transmission media can be classified as **guided or unguided**.
- ✧ In both cases, communication is in the form of **electromagnetic waves**.
 - With guided media, the waves are **guided along a solid medium**, such as **copper twisted pair, copper coaxial cable, and optical fiber**.
- ✧ The **atmosphere** and **outer space** are examples of **unguided media** that provide a means of transmitting electromagnetic signals but do not guide them; this form of transmission is usually referred to as **wireless transmission**.

Contd.



Design Factors

- A key concern is data rate and distance.
 - the greater the data rate and distance, the better.
- A number of design factors relating to the transmission medium and to the signal determine the data rate and distance:
 - Bandwidth
 - Transmission impairments
 - Interference
 - A Number of receivers

Contd.

- **Bandwidth:-**
 - All other factors remaining constant, the greater the bandwidth of a signal, the higher the data rate that can be achieved.
- **Transmission impairments:-**
 - Impairments, such as attenuation, limit the distance.
 - For guided media, **twisted pair** generally suffer more impairment than **coaxial cable**, which in turn suffers more than **optical fiber**.

Contd.

■ Interference:-

- Interference from competing signals in overlapping frequency bands can distort or wipe out a signal.
- particular concern for unguided media, but it is also a problem with guided media.
- For guided media, interference can be caused by emanations from nearby cables.
- For example, twisted pair are often bundled together, and conduits often carry multiple cables. Proper shielding of a guided medium can minimize this problem.

Contd.

■ A Number of receivers:-

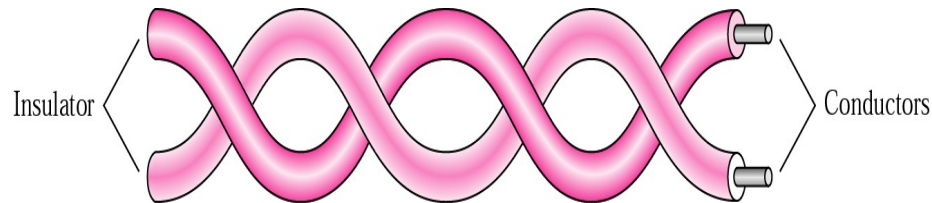
- A guided medium can be used to construct a point-to-point link or a shared link with multiple attachments.
- In a shared link , each attachment introduces some attenuation and distortion on the line, limiting distance and/or data rate.

GUIDED TRANSMISSION MEDIA

- ✈ include twisted-pair cable, coaxial cable, and fiber-optic cable.
- ✈ A signal traveling along any of these media is directed and contained by the physical limits of the medium.
- ✈ Twisted-pair and coaxial cable use metallic (copper) conductors that accept and transport signals in the form of electric current.
- ✈ Optical fiber is a cable that accepts and transports signals in the form of light.

Twisted-Pair Cable

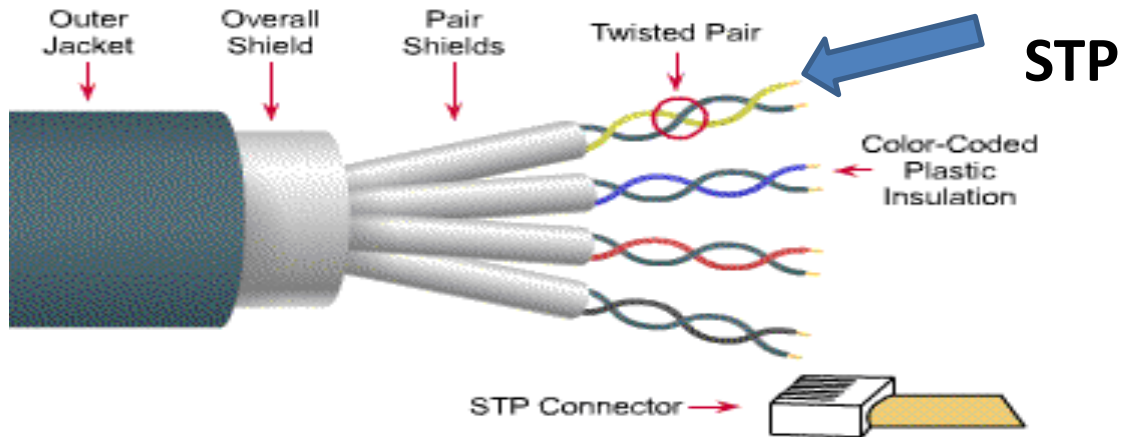
- ✎ consists of two conductors (normally copper), each with its own plastic insulation, twisted together, as shown in the figure below



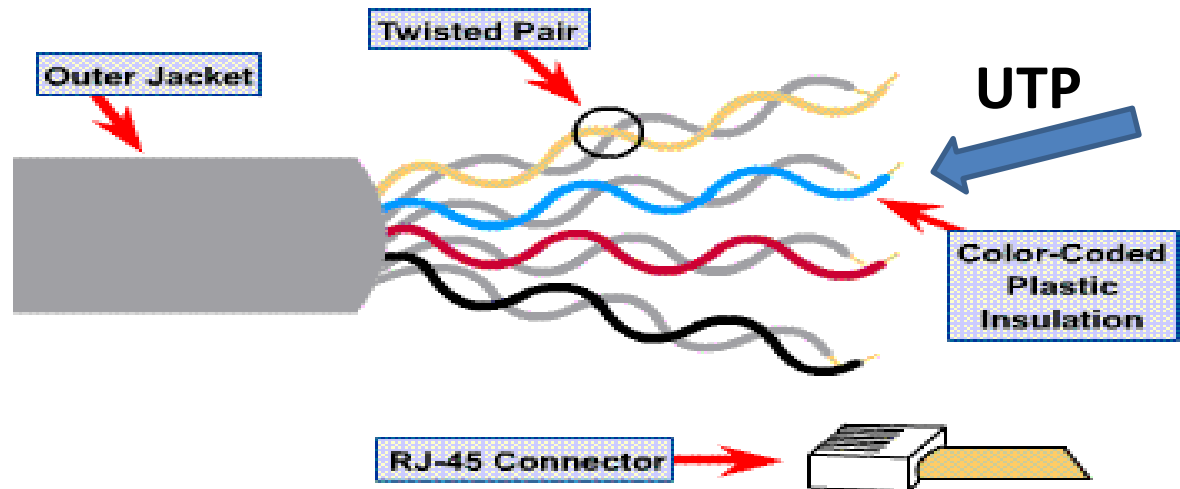
- ✎ One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference.
- ✎ The receiver uses the difference between the two.
- ✎ In addition to the signal sent by the sender on one of the wires, interference (noise) and crosstalk may affect both wires and create unwanted signals.
- ✎ the **number of twists per unit of length** (e.g., inch) has some effect on the **quality of the cable**.

Unshielded Versus Shielded Twisted-Pair Cable

- ✉ The most common twisted-pair cable used in communications is referred to as unshielded twisted-pair (UTP).
- ✉ IBM has also produced a version of twisted-pair cable for its use called shielded twisted-pair (STP).
- ✉ STP cable **has a metal foil or braided mesh covering** that encases each pair of insulated conductors. Although metal casing improves the quality of cable by **preventing the penetration of noise or crosstalk**, it is bulkier and more expensive.



- Speed and throughput: 10-100 Mbps
- Cost per node: Moderately expensive
- Media and connector size: Medium to Large
- Maximum cable length: 100m (short)



- Speed and throughput: 10-100 Mbps
- Cost per node: Least Expensive
- Media and connector size: Small
- Maximum cable length: 100m (short)

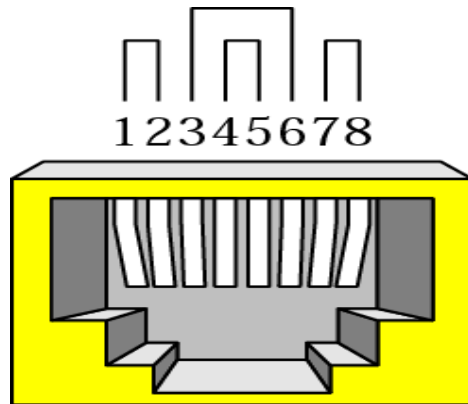
UTP Categories

- The Electronic Industries Association (EIA) has developed standards to classify unshielded twisted-pair cable into **seven categories**.
- **Categories are determined by cable quality, with 1 as the lowest and 7 as the highest.**
- Each EIA category is suitable for specific uses.

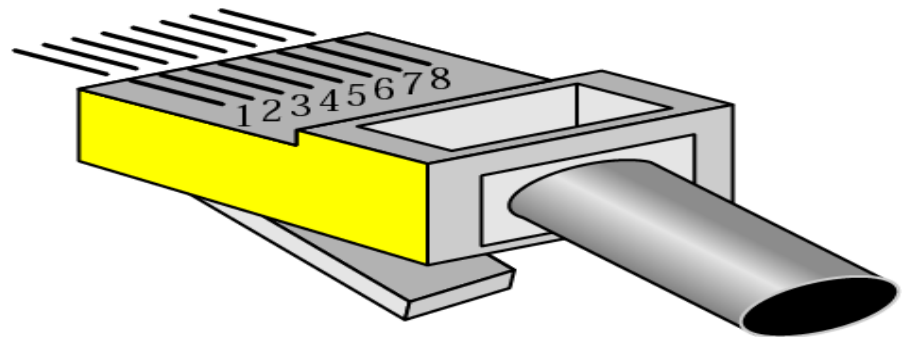
<i>Category</i>	<i>Specification</i>	<i>Data Rate (Mbps)</i>	<i>Use</i>
1	Unshielded twisted-pair used in telephone	< 0.1	Telephone
2	Unshielded twisted-pair originally used in T-lines	2	T-llines
3	Improved CAT 2 used in LANs	10	LANs
4	Improved CAT 3 used in Token Ring networks	20	LANs
5	Cable wire is normally 24 AWG with a jacket and outside sheath	100	LANs
SE	An extension to category 5 that includes extra features to minimize the crosstalk and electromagnetic interference	125	LANs
6	A new category with matched components coming from the same manufacturer. The cable must be tested at a 200-Mbps data rate.	200	LANs
7	Sometimes called SSTP (shielded screen twisted-pair). Each pair is individually wrapped in a helical metallic foil followed by a metallic foil shield in addition to the outside sheath. The shield decreases the effect of crosstalk: and increases the data rate.	600	LANs

Connectors

- The most common UTP connector is **RJ45** (RJ stands for **registered jack**). The RJ45 is a **keyed connector**, meaning the connector can be inserted in only one way.



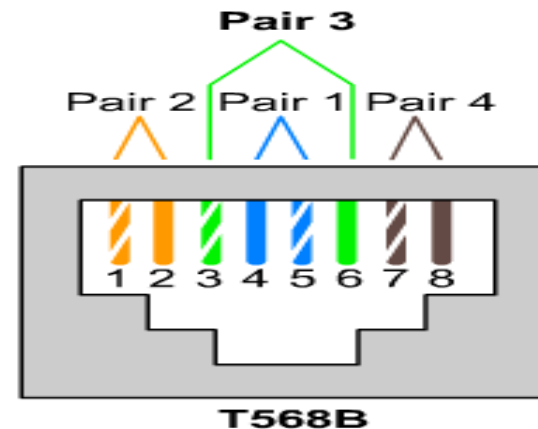
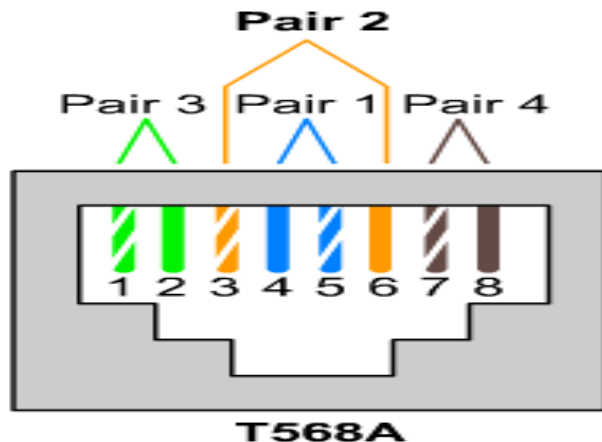
RJ-45 Female



RJ-45 Male

RJ-45 connector

- A **straight-through cable** has T568B on both ends.
- A **crossover (or cross-connect)** cable has T568B on one end and T568A on the other.
- A **console cable** had T568B on one end and **reverse T568B** on the other, which is why it is also called a **rollover cable**.



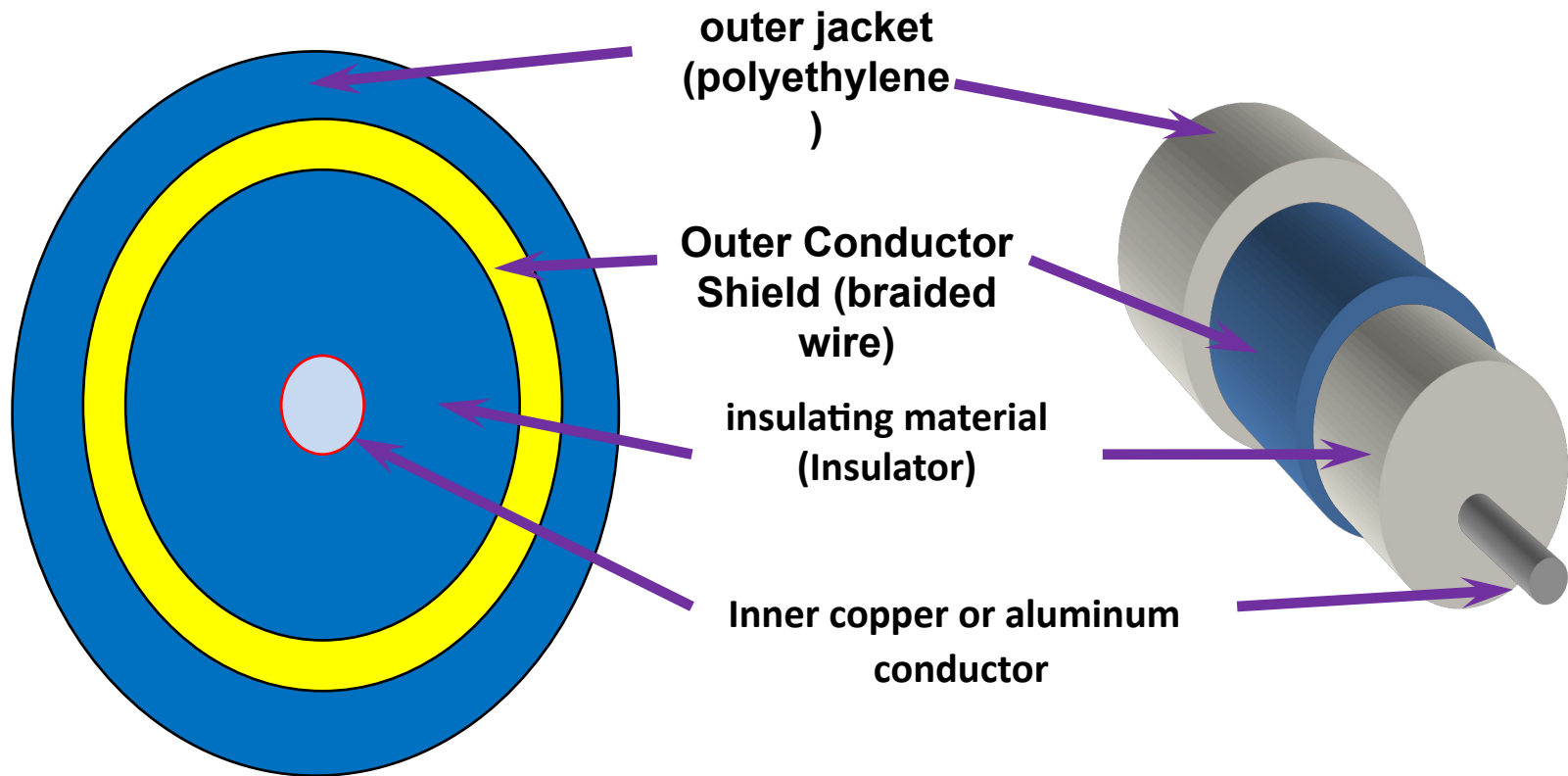
When to use ?

- Use **straight-through** cables for the following cabling:
 - Switch to router
 - Switch to PC or server
 - Hub to PC or server
- Use **rollover** cable
 - used to connect a computer terminal to a router's console port.
- Use **crossover** cables for the following cabling:
 - Switch to switch
 - Switch to hub
 - Hub to hub
 - Router to router
 - PC to PC
 - Router to PC

Twisted-pair cable Applications

- Within buildings
- Most common medium
- Twisted-pair cables are used in **telephones lines** to provide voice and data channels.
- The **DSL lines** that are used by the telephone companies to provide high data rate connections also use the high-bandwidth capability of unshielded twisted-pair cables.
- **Local area networks**, such as **10Base-T** and **100Base-T**, also used UTP cables.

Coaxial Cable



Coaxial Cable

- ✎ Coaxial cable (or *coax*) carries signals of *higher frequency ranges than those in twisted pair cable*, in part because the two media are constructed quite differently.
- ✎ Instead of having two wires, coax has a central core conductor of **solid or stranded wire (usually copper)** enclosed in an insulating sheath, which is, in turn, encased in an **outer conductor of metal foil, braid**, or a combination of the two.
- ✎ **The outer metallic wrapping** serves both as a **shield against noise** and as the **second conductor**, which completes the circuit. This outer conductor is also enclosed in an insulating sheath, and the whole cable is protected by a plastic cover

Contd.

- At one time, coaxial cable was the most widely used network cabling. There were a couple of reasons for coaxial cable's wide usage:
 - it was relatively inexpensive
 - it was light, flexible, and easy to work with.

Contd.

- **Shielding protects** transmitted data by absorbing stray electronic signals, called noise, so that they do not get onto the cable and distort the data.
- Cable that contains **one layer of foil insulation** and **one layer of braided metal shielding** is referred to as **dual shielded**.
- For environments that are subject to higher interference, **quad shielding is available**.
- Quad shielding consists of two layers of foil insulation and two layers of braided metal shielding.

Types of Coaxial Cable

1. Thin (thinnet) cable

2. Thick (thick net) cable

1. Thinnet(10Base2):-

- is a flexible coaxial cable about **0.64 centimeters (0.25 inches)** thick.
- distance of up to approximately 185 meters
 - Because this type of coaxial cable is flexible and easy to work with
 - it can be used in almost any type of network installation.

Types of Coaxial Cable:

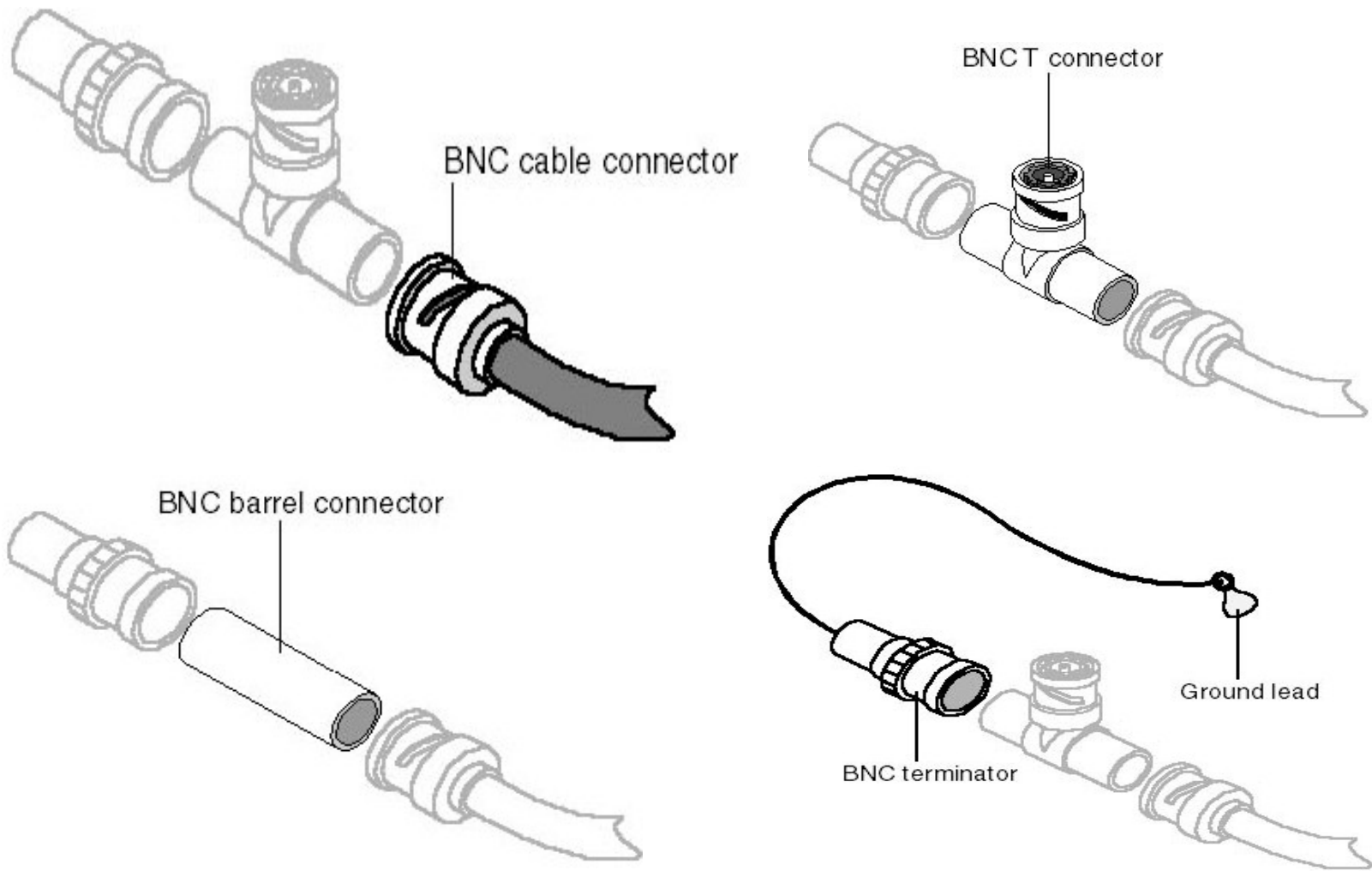
Thicknet(10Base5)

- is a relatively rigid coaxial cable about **1.27** centimeters (0.5 inches) in diameter.
- ✎ is sometimes referred to as **Standard Ethernet** because it was the first type of cable used with the popular network architecture Ethernet.
- ✎ **is thicker than a thinnet** cable core.
- ✎ **thick net can carry signals farther than thinnet cable.** Thicknet cable can carry a signal for 500 meters (about 1640 feet).
- ✎ **used as a backbone to connect several smaller thin net-based networks**

Thin net vs. Thick net

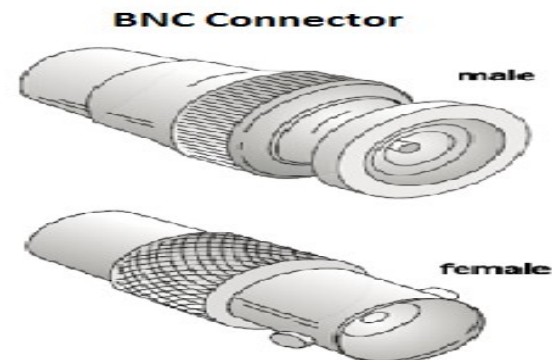
- ✎ The thicker the cable, the more difficult it is to work with.
- ✎ Thin cable is flexible, easy to install, and relatively inexpensive.
- ✎ Thick cable does not bend easily and is, therefore, harder to install.
- ✎ Thick cable is more expensive than thin cable, but will carry a signal farther.

Connector



BNC connector

- BNC T connector is used to connect the end of the cable to a device, such as a TV set.
- The BNC T connector is used in Ethernet networks to branch out to a connection to a computer or other device.
- The BNC terminator is used at the end of the cable to prevent the reflection of the signal
- BNC barrel connector is used to join two lengths of thinnet cable to make one longer length.



Categories of coaxial cables

Category	Impedance	Use
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

Coax advantages

- It is less susceptible to interference than twisted-pair cable.
- Transmit data for greater distances than is possible with less expensive cabling.
- Transmit voice, video, and data
- Offer a familiar technology with reasonable data security.

Coax disadvantages

- There are many types of coax cables, each suited for one, or at most, a small number of LAN systems.
- Due to its high metallic content, coax cable is usually more expensive than other cable types.

Coaxial Cable Applications

- Most versatile medium
- Television distribution
 - Ariel to TV
 - Cable TV
- Long distance telephone transmission
 - Can carry 10,000 voice calls simultaneously
 - Being replaced by fiber optic
- Short distance computer systems links
- Local area networks

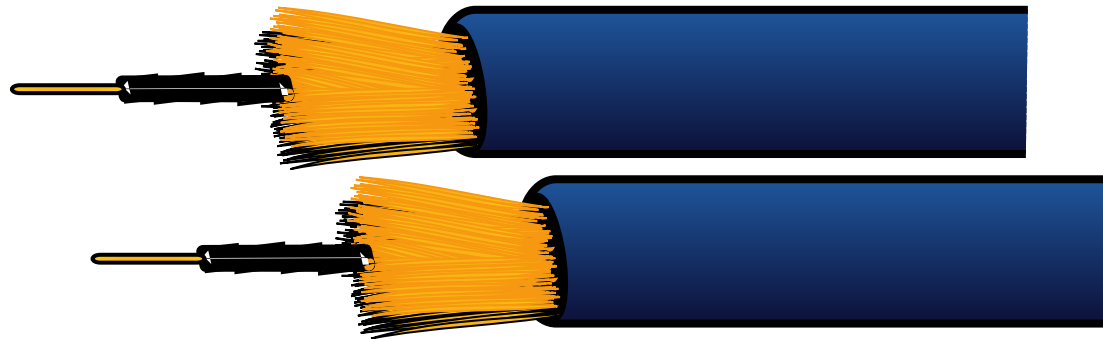
Fiber-Optic Cable

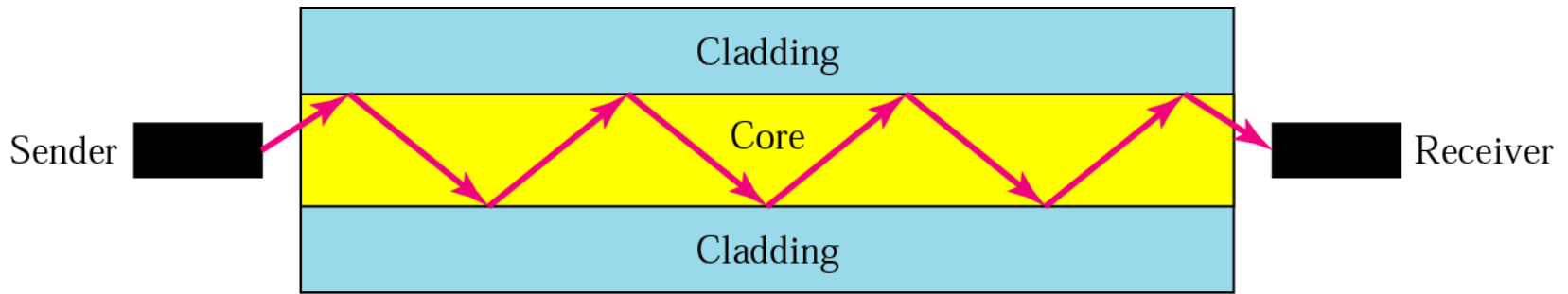
- ✧ is made of **glass or plastic** and transmits **signals in the form of light**.
- ✧ consists of a signal-carrying glass core of **5 to 100 microns** in diameter (a sheet of paper is about 25 microns thick and a human hair about 75 microns thick), surrounded by a layer of pure silica called **cladding**, which prevents light from escaping.
- ✧ Surrounding the cladding are **protective layers** of acrylic plastic coating, **Kevlar fibers** for additional strength, and a PVC (polyvinyl chloride) jacket (usually colored a distinctive orange).

Contd.

✈ Network components use **LED or laser diodes** to convert electrical signals into light pulses for transmission on fiber-optic cables.

✈ An **optical detector** is used to convert the light pulses back into electrical signals.





- ✎ Optical fibers use reflection to guide light through a channel.
- ✎ A glass or plastic core is surrounded by a cladding of less dense glass or plastic.
- ✎ The difference in density of the two materials must be such that a beam of light moving through the core is reflected off the cladding instead of being refracted into it (cladding)

Optical Fiber - Applications

- Long-haul trunks
- Metropolitan trunks
- Rural exchange trunks
- Subscriber loops
- LANs

Advantages of Optical Fiber

- ✉ The major advantages offered by fiber-optic cable over twisted-pair and coaxial cable are **noise resistance**, **less signal attenuation**, and **higher bandwidth**.
1. **Noise Resistance:** Because fiber-optic transmission uses light rather than electricity, noise is not a factor. External light, the only possible interference, is blocked from the channel by the outer jacket.
 2. **Less signal attenuation:-** Fiber-optic transmission distance is significantly greater than that of other guided media. A signal can run for miles without requiring regeneration.
 3. **Higher bandwidth:-** Currently, data rates and bandwidth utilization over fiber-optic cable are limited not by the medium but by the signal generation and reception technology available.

Disadvantages of Optical Fiber

- The main disadvantages of fiber optics are **cost, installation/maintenance, and fragility.**
- 1. **Cost:-** Fiber-optic cable is expensive. Also, a laser light source can cost thousands of dollars, compared to hundreds of dollars for electrical signal generators.
- 2. **Installation/maintenance**
- 3. **Fragility:-** Glass fiber is more easily broken than wire, making it less useful for applications where hardware portability is required.

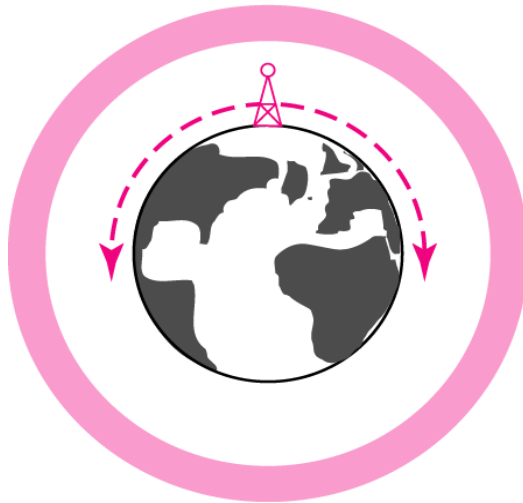
Unguided (Wireless) Media

- ✉ Unguided media transport electromagnetic waves without using any physical conductor.
- ✉ This type of communication is often referred to as wireless communication.
- ✉ Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.

Unguided (Wireless) Media

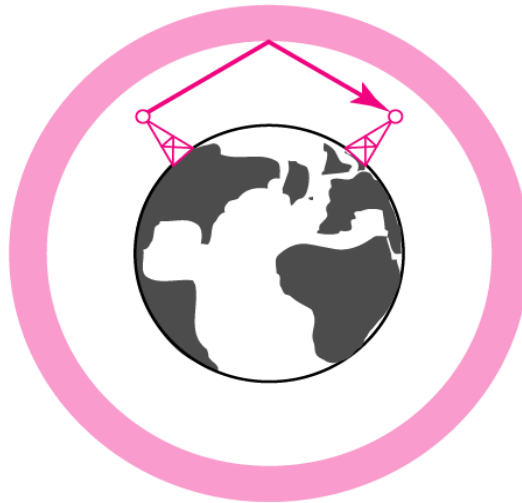
✉ Unguided signals can travel from the source to destination in several ways: **ground propagation**, **sky propagation**, and **line-of-sight propagation**.

Ionosphere



Ground propagation
(below 2 MHz)

Ionosphere



Sky propagation
(2–30 MHz)

Ionosphere



Line-of-sight propagation
(above 30 MHz)

Contd.

1. In ground propagation

- radio waves travel through the lowest portion of the atmosphere, hugging the earth.
- These low-frequency signals emanate in all directions from the transmitting antenna and follow the curvature of the planet.
- Distance depends on the amount of power in the signal:
The greater the power, the greater the distance.

Contd.

2.In sky propagation

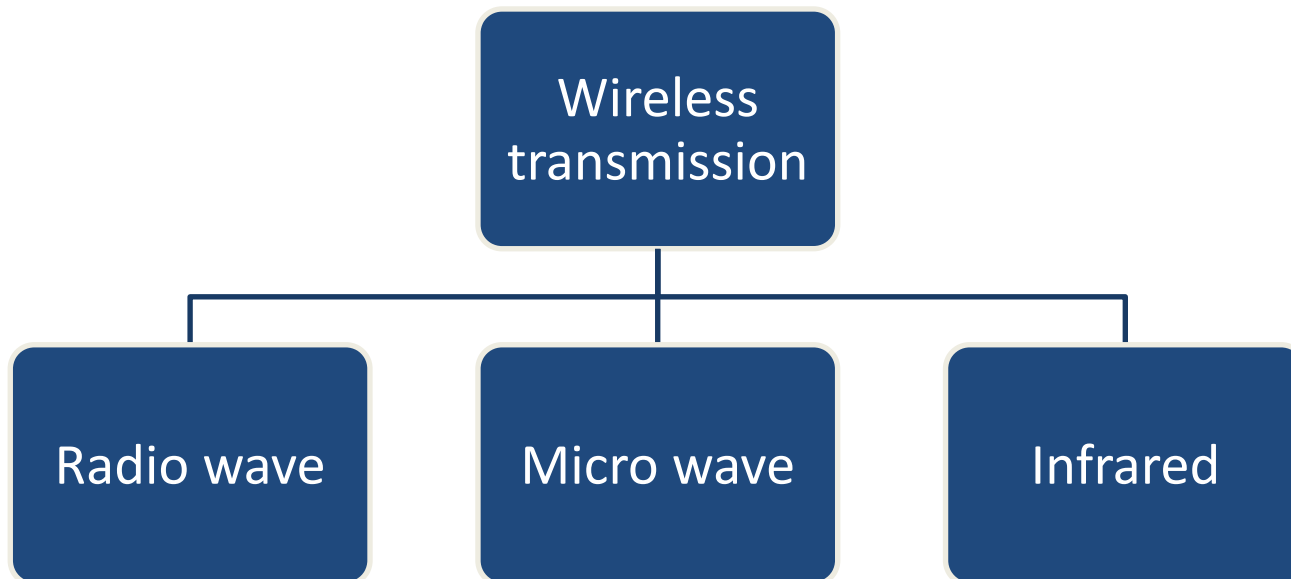
- higher-frequency radio waves radiate upward into the ionosphere (the layer of atmosphere where particles exist as ions) where they are reflected back to earth.
- This type of transmission allows for greater distances with lower output power.

Contd.

3. In line-of-sight propagation

- very high-frequency signals are transmitted in straight lines directly from antenna to antenna.
- Antennas must be directional, facing each other, and either tall enough or close enough together not to be affected by the curvature of the earth.

Wireless Transmission Types



Radio Waves

- waves that are transmitted in all the directions of free space.
- are omnidirectional, i.e., the signals are propagated in all the directions.
- The range in frequencies of radio waves is from 3Khz to 1 khz.
- In the case of radio waves, the sending and receiving antenna are not aligned, i.e., the wave sent by the sending antenna can be received by any receiving antenna.
- An example of the radio wave is **FM radio**.

Microwaves

- **Frequency range:** The frequency range of terrestrial microwave is from 4-6 GHz to 21-23 GHz.
- **Bandwidth:** It supports the bandwidth from 1 to 10 Mbps.
- **Short distance:** It is inexpensive for short distance.
- **Long distance:** It is expensive as it requires a higher tower for a longer distance.
- **Attenuation:** It is affected by environmental conditions and antenna size.
- 🔊 Microwaves are unidirectional. When an antenna transmits microwave waves, they can be narrowly focused. (the sending and receiving antennas need to be aligned /see each other).

Infrared

- ✧ An infrared transmission is a wireless technology used for communication over short ranges.
- ✧ The frequency of the infrared is in the range from 300 GHz to 400 THz.
- ✧ It is used for short-range communication such as data transfer between two cell phones, TV remote operation, data transfer between a computer and cell phone resides in the same closed area.

